Stylus Color Line Eraser Backgrounds Undo Redo Pages Previous Next Erase Board Web Documents Show Desktop OpenBoard

Ques: Matrix Chain Multiplication (MCM)

$A_{m \times n} \times B_{p \times q} = C_{m \times q}$ & $n == p$

$(A)(B)(C)(D)$

$A(B \cdot C) = (AB) \cdot C$

$ABCD$

- $A(BCD)$
  - $A[(BC)D]$
  - $A[B \cdot (CD)]$
- $(AB) \cdot (CD)$
- $(ABC)D$
  - $[(AB) \cdot C]D$
  - $[A(BC)]D$

Stylus Color Line Eraser Backgrounds Undo Redo Pages Previous Next Erase Board Web Documents Show Desktop OpenBoard

Minimum no. of multiplications

$A_{2\times3} \cdot B_{3\times4} \cdot C_{4\times5}$

$(AB)\cdot C$

$$\underset{A}{\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}_{2\times3}} \cdot \underset{B}{\begin{bmatrix} g & h & i & j \\ k & l & m & n \\ o & p & q & r \end{bmatrix}_{3\times4}} = \underset{AB}{\begin{bmatrix} s & t & u & v \\ w & x & y & z \end{bmatrix}}$$

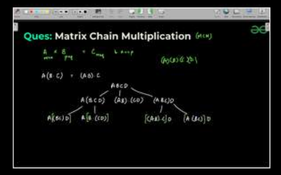
total multiplications $= 2 \times 3 \times 4$

In general,

no. of multiplications

in $A_{m\times n} \cdot B_{n\times p}$

is $\boxed{m \times n \times p}$



Stylus Color Line Eraser Backgrounds Undo Redo Pages Previous Next Erase Board Web Documents Show Desktop OpenBoard

Minimum no. of multiplications

$$A_{2\times3} \cdot B_{3\times4} \cdot C_{4\times5}$$

$(AB)\cdot C$

$2\times3\times4$

$+$

$2\times4\times5$

$AB_{2\times4} \cdot C_{4\times5}$

$\boxed{64}$

✓

$A\cdot(BC)$

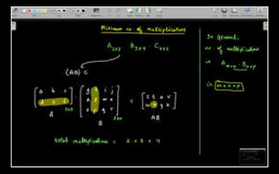
$3\times4\times5$

$+$

$2\times3\times5$

$A_{2\times3} \cdot BC_{3\times5}$

$\boxed{90}$



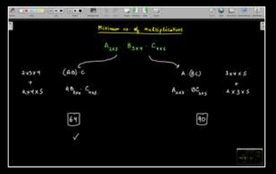
Stylus Color Line Eraser Backgrounds Undo Redo Pages Previous Next Erase Board Web Documents Show Desktop OpenBoard

Input Format

$$\{\{2,3\},\ \{3,4\},\ \{4,5\},\ \{5\times 6\},\ \{6,7\}\}$$

or

$$\{2,\ 3,\ 4,\ 5,\ 6,\ 7\}$$



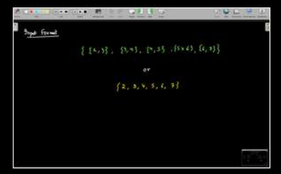
Stylus Color Line Eraser Backgrounds Undo Redo Pages Previous Next Erase Board Web Documents Show Desktop OpenBoard

Code

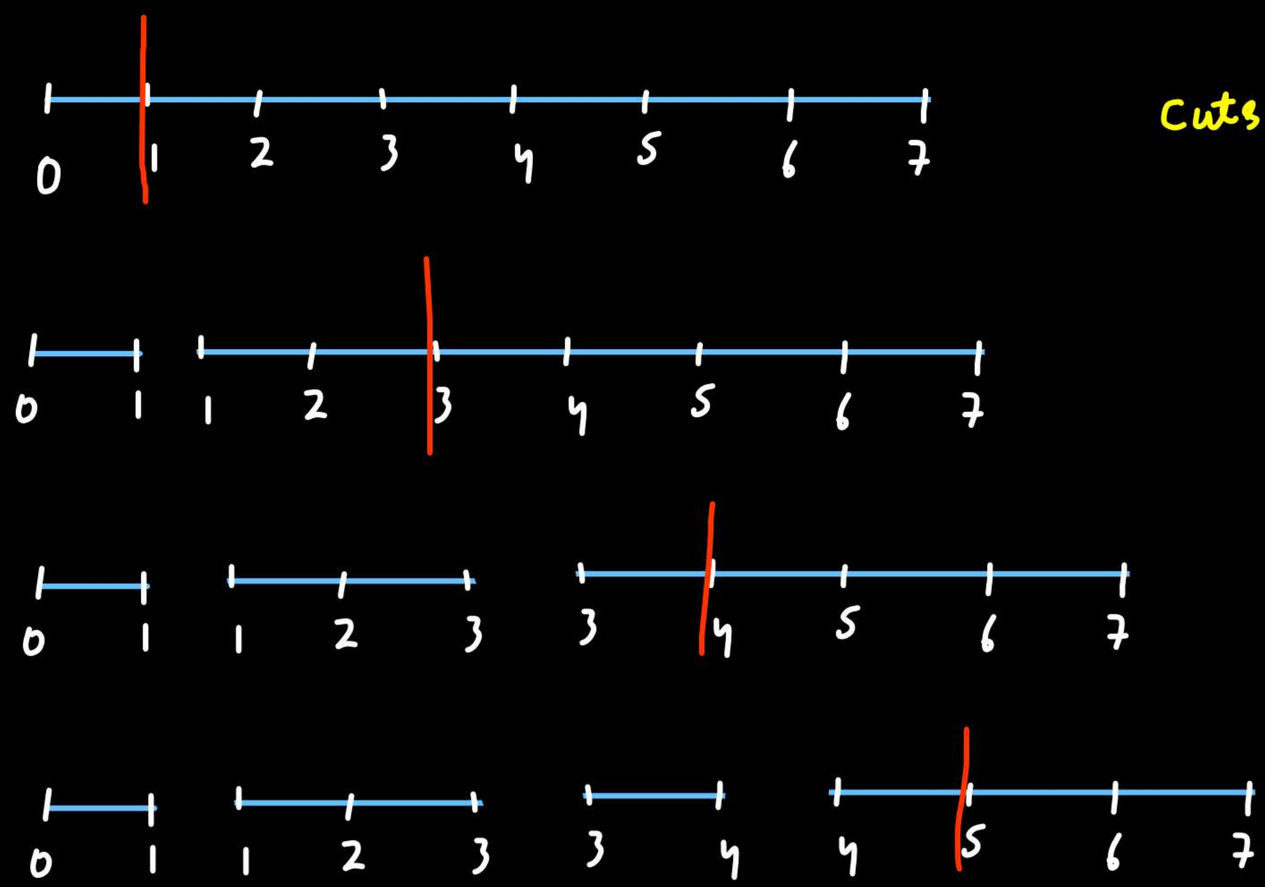
$$\begin{array}{cccccc} i & & k & k+1 & j & \\ \end{array}$$
$$arr = \{ 2, 3, 4 \mid 5, 6, 7 \}$$
$$\begin{array}{cccccc} A & B & C & D & E & \end{array}$$

$T.C. = O(n^3)$

```
fun(i,j){
    if(j == i)  return 0;
    for( k = i to j-1)
        cost = arr[i] × arr[j+1] × arr[k+1]
        multiplications = fun(i,k) + fun(k+1,j) + cost
        minCost = min(multiplications, mincost)
    }
    return mincost
}
```



Ques: Minimum Cost to cut a stick



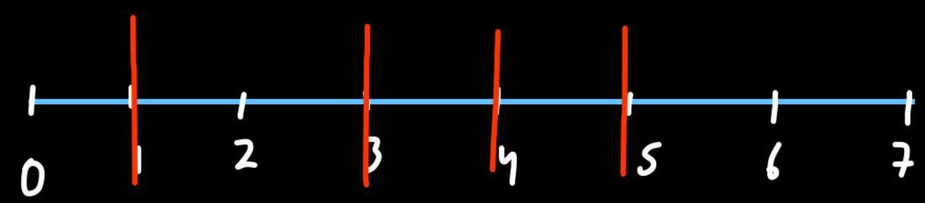
cuts = {1, 3, 4, 5}

n = 7

cost = 7 + 6 + 4 + 3 = 20

Stylus Color Line Eraser Backgrounds Undo Redo Pages Previous Next Erase Board Web Documents Show Desktop OpenBoard

Ques: Minimum Cost to cut a stick



cuts = {1, 3, 4, 5}

n = 7

cost = 7 + 4 + 3 + 2 = 16

Stylus Color Line Eraser Backgrounds Undo Redo Pages Previous Next Erase Board Web Documents Show Desktop OpenBoard

Code

cuts = { 5, 4, 3, 1 }

n = 7

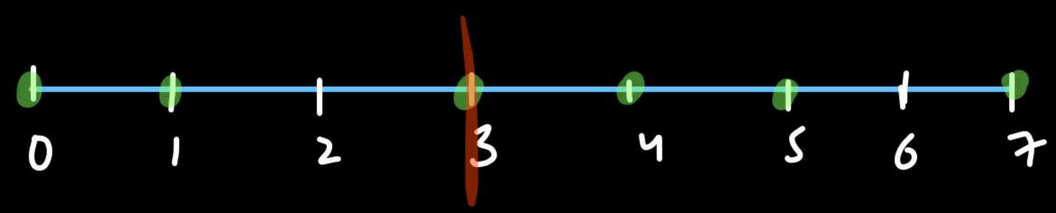
arr = { 0, 1, 3, 4, 5, 7 }

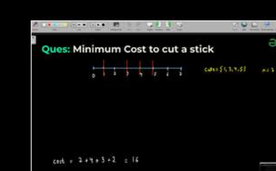
i k j

{ 0, 1, 3 } { 3, 4, 5, 7 }

i j i k j

{ 3, 4 } { 4, 5, 7 }




```
for(K = i+1 to j-1){
    cost = (arr[j] - arr[i]) + fun(i,K) + fun(K,j)
    minCost = min(cost, mincost)
}
```